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Medium-term feeder load forecasting and boosting peak accuracy prediction using the PWP-XGBoost model

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ABSTRACT

Medium-term feeder load forecasting plays a pivotal role in efficiently operating and planning electrical distribution systems. It provides valuable insights into future electricity demand trends, enabling utilities to make informed decisions regarding infrastructure upgrades, resource allocation, and energy management strategies. However, few studies have been done on medium-term load forecasting for the distribution network's operational planning at the medium voltage level. Moreover, conventional load forecasting techniques mainly consider the impact of limited external factors, which is typically challenging to forecast accurately. In this work, multiple influential features have been utilized for accurate medium-term load prediction. Accurate load forecasting is paramount for efficient operation and planning in power systems. This study proposes a novel approach for medium-term feeder load forecasting that enhances peak accuracy using the Prominence-guided Weighted Peaks (PWP) in conjunction with the eXtreme Gradient Boosting (XGBoost) model. To evaluate the performance of the proposed model, we conduct experiments using real-world load data from a distribution feeder. Comparative analysis with traditional forecasting methods demonstrates the superior accuracy of the PWP-XGBoost model, particularly in predicting peak loads. Enhanced peak accuracy is crucial for utilities to effectively manage peak demand, optimize resource allocation, and ensure grid stability.

1. Introduction

Global energy consumption is increasing with the rise in population and economic prosperity. Energy is the lifeblood of modern civilization, which powers everything from our homes to industry and technological advancements. Efficient energy management is crucial, given the growing concern over environmental sustainability and the limited availability of fuels. Energy management entails monitoring, controlling, and optimizing energy resource use, thus achieving goals like reducing energy consumption and increasing cleaner energy usage. Energy prediction, or load forecasting, is a tool for effective energy management. Load forecasting is critical for ensuring the reliable and efficient operation of transmission and distribution systems within the energy industry. It has a significant role in grid operation in the power industry. Accurate load forecasts enable utilities to allocate resources effectively, optimize energy generation and distribution, and plan future infrastructure upgrades. There are various reasons why load forecasting is necessary. It facilitates

- Grid Operations and Planning: Load forecasts assist grid operators in optimizing power generation and load balancing, ensuring that the electricity supply matches the expected demand. Moreover, it helps determine the optimal utilization of existing assets and planning future infrastructure developments.
- Energy Security: Accurate load predictions in extreme weather conditions ensure a stable and secure energy supply, reducing the risk of blackouts and energy shortages, which can have severe economic and societal repercussions.
- Renewable Energy Integration: We strive to adopt cleaner energy sources such as solar and wind; load forecasting will help us perform accurate load flow estimation and hence understand the impacts of the renewable resource integration on the network, such as peak load and reverse power flow, resource allocation, and balance supply and demand.
- Energy Purchasing and Trading: Load forecasting enables the network operator to estimate the required energy from the wholesale market, enabling energy traders, the electricity suppliers, and traders to make informed decisions about buying or selling electricity, which leads to optimized, cost-effective transactions

Load forecasting has been a concern since the early days of electricity generation and distribution. In the late 19th century, as electric power

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systems were developing, load forecasting was rudimentary but crucial for ensuring adequate supply. Load forecasting relied heavily on manual methods. Operators and engineers would record historical load data, seasonal patterns, and weather conditions to predict the near future. In the mid-20th century, statistical techniques such as time series analysis, regression analysis, and moving averages gained popularity for load forecasting. These methods allowed for more systematic and data-driven predictions. With the advent of computers in the mid-20th century, load forecasting began to benefit from computational power. Computer-based models and simulations enable more sophisticated forecasting approaches. Weather conditions significantly impact electricity demand, especially for heating and cooling. Integrating weather data into load forecasting models has become common in capturing this influence accurately. The power industry has increasingly adopted machine learning and artificial intelligence (AI) techniques for load forecasting in recent decades.

Based on the forecasting horizon, load forecasting can be broadly classified into three categories: short-term load forecasting, medium-term load forecasting, and long-term load forecasting. Short-term forecasts typically span from one hour to one week, medium-term forecasts usually range from a week to a year, and long-term forecasts extend beyond a year. Short-term load forecasting (STLF) [1–4] is crucial for managing and optimizing power grids' day-to-day operations, enabling real-time adjustments to meet immediate demand fluctuations. Medium-term load forecasting [5–7] aids in resource planning, helping utilities decide on capacity expansion and resource allocation over several months to a year. Long-term load forecasting [8,9] is pivotal in more comprehensive strategic planning, guiding investments in infrastructure and multi-year energy policy decisions.

Medium-term load forecasting is crucial for grid operators and planners to manage the electricity grid efficiently. It involves predicting electricity demand over a time horizon ranging from a few days to several months, typically between one day and one year. This forecasting helps grid operators make informed decisions and optimize the operation of the power system. It aids in making informed decisions about resource allocation, infrastructure planning, market operations, and emergency preparedness. Generally, as the horizon of energy prediction increases, forecast accuracy reduces. Different methods have been identified for short-term load forecasting [2,10–13]. We can observe significantly less research for medium-term load forecasting, especially for one year ahead.

Prediction is solely based on the past values of the load profile in univariate load forecasting. In contrast, multivariate load forecasting depends on several features that can help with better prediction. Researchers have extensively explored various AI techniques to optimize load forecasting, addressing the crucial challenges in energy resource allocation and grid stability. In univariate load forecasting, prediction is solely based on the past values of the load profile. In contrast, multivariate load forecasting depends on several features that can help with a better prognosis. Hence, we have considered multivariate load forecasting rather than univariate load forecasting. Traditional time series models like ARIMA [1] and its derivatives are mainly designed for univariate load forecasting; they focus on the short-term forecasting horizon. Support Vector Machines (SVMs) have shown promise in handling nonlinear forecasting tasks [2]. Ensemble models like Random Forest and XGBoost are showing higher prediction accuracy for multivariate load forecasting. Artificial Neural Networks (ANNs), particularly multi-layer feedforward neural networks, are widely adopted to capture complex data patterns. However, they cannot learn the temporal dependencies of data. The rise of RNN and LSTM overcame this drawback of deep learning [14]. None of these models focus on medium-term predictions, i.e., 1 year ahead. The recent work, Dynamic artificial neural network model [15] performs one year ahead of load prediction. However, the complex nature of neural networks limits their practical application to deployment and also restricts their usage when data is less. The paper [16]highlights the requirement of scalability of the models and its deployment. There are models for predicting daily peak load [17], but predicting annual peak load requires much further modification. Even hybrid models like XGBoost-LSTM [18], wherein the model separately forecasts the energy profile and peak load value and then combines them for improved accuracy, concentrate on projecting short-term load, not medium-term. Load forecasting is a critical component of managing smart grid systems, especially prosumer-centric and self-sustained ones [19,20]. Accurate load forecasting enables better prediction of load demand peaks, which is essential for demand-supply balance, peak load management, energy storage optimization, cost efficiency, and renewable integration.

Predicting next year's peak load value is challenging. This paper aimed to achieve a one-year-ahead load prediction by considering multiple features and accurately predicting the following year's peak load value. The primary contributions of this research work are as follows:

- In this work, one year-ahead load prediction at the feeder level has been performed. Multiple influential features like temperature, humidity, the number of customers KVA, the number of substations, holidays, and all other relevant temporal data like hours, weeks, months, and quarters of the year are considered for load forecasting.
- Multivariate analysis has been performed to understand the effect of each feature on load value before machine learning modeling. Also, a technique has been developed after modeling to understand how each feature helped load forecasting.
- Predicting peak accuracy in a year is one of the challenges faced by the distribution and planning team. This paper introduced a novel technique into the machine learning model, XGBoost, to enhance peak prediction accuracy. A prominence-guided weighted peak prediction model has been proposed to improve peak accuracy along with our half-hour-resolution medium-term load forecasting.

2. Methodology

The load forecasting methodology involves data collection, preprocessing, analysis, and identifying the right features, followed by model selection, training, and testing. Along with load forecasting, we also focused on improving the accuracy of peak load prediction. It involves identifying the peaks and the corresponding peak hours and months and incorporating these while optimizing the model. The block diagram in Fig. 1 illustrates the whole load forecasting process. The Block Diagram unveils the stages of analysis, training, and validation for robust model development for medium-term load forecasting. The PWP - XGBoost model also uses the information of peak values for optimizing the peak load prediction. Once the year ahead peak value is available, the distribution planning team can use the information for planning their future requirements.

2.1. Data collection and processing

2.1.1. Topology

This study aims to forecast the load at the individual feeder Loads, as shown in Fig. 2, by building customized predictive models. In this study, we have considered a part of the Dubai distribution network with one primary substation and downstream feeders. The model will also consider the below features as inputs: 1. Actual Main Substation Load. 2. Actual Individual Feeder Load. 3. Connected Installed Capacity. 4. Network Connectivity Model. 5. Number of Customer Connections. 6. Temperature. 7. Seasons Start and End Dates. Necessary data cleansing will be done to validate the data and ensure their quality. The process for data cleaning is explained in the next section.

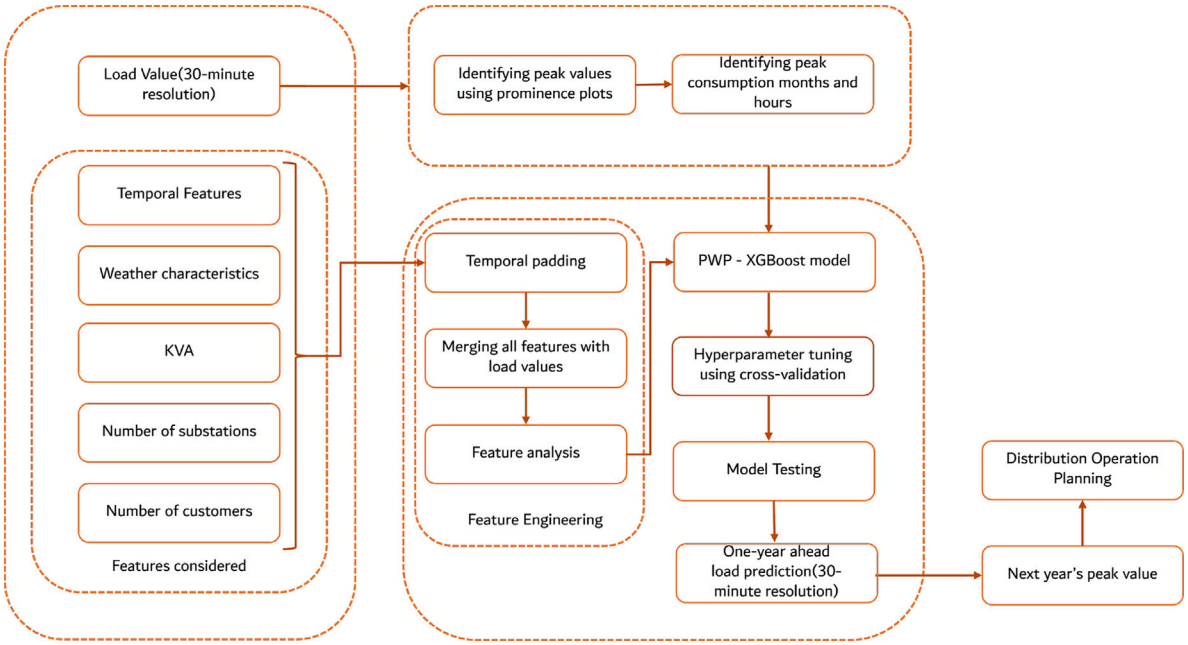


Fig. 1. Load forecasting using PWP - XGBoost model.

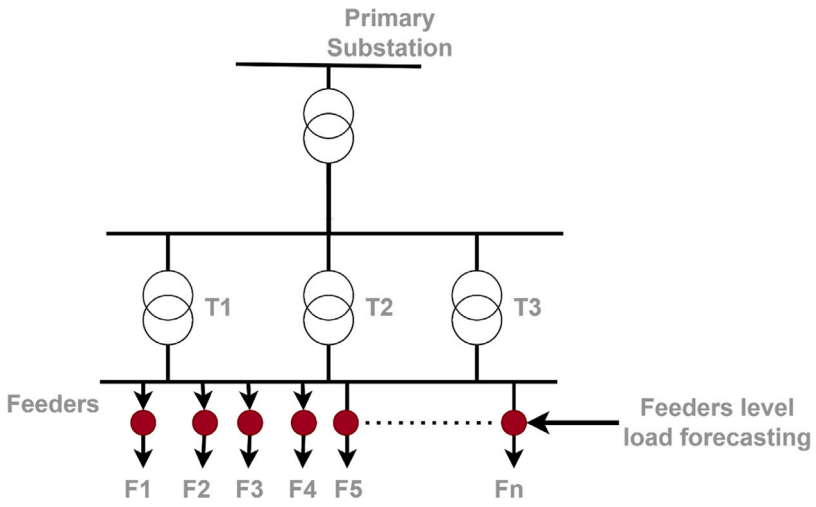


Fig. 2. Distribution network.

2.1.2. Data pre-processing

This work uses a real feeder load value of 5 feeders within the Dubai distribution network with data recorded at 30-minute data resolution. The dataset size captured from each feeder consists of 102,340 data points. The included features are temperature and humidity from the National Center of Meteorology & Seismology (NCMS) along with the number of customers, KVA, and the number of substations. We used these features to develop a machine-learning model for estimating medium-term loads. All of these features are at different resolutions, which are converted into a single resolution. As the load value is in a 30-minute resolution, we converted all other features to a 30-minute resolution. We attribute values for converting values from lower resolution to higher resolution. We used padding where the previously observed last value will fill the missing values.

2.1.3. Data analysis and feature selection

Fig. 3 depicts the load profile for feeder 1, which we have selected as a visual example for this paper. We can observe a seasonal nature in the load profile, which helps our machine learning models learn the intricate pattern effectively, thus facilitating accurate prediction and pattern recognition. Other feeders also follow a similar load profile pattern.

Feature selection: We considered features that will affect forecasting using domain expertise and temporal aspects that capture intricate patterns of load demand over time. We have also considered features like Connected Installed Capacity, and Customer Connections. We considered the climatic features like temperature and humidity affecting load consumption. Temporal features like a year, month, day, hour, day of the year, week quarter, holidays, and week date are significant in capturing the pattern of time-related load. The importance of these features and the correlation between load values is discussed below.

A box plot shown in Fig. 4(a) demonstrates how energy load varies during every month of the year. Visualizing each month's data distribution and central tendency of load profiles supports our feature selection process. The x-axis of the plot is divided into 12 sections, each box corresponding to a month of the year. The box in the middle of the plot represents the interquartile range (IQR), which is the range between the first quartile (Q1) and the third quartile (Q3) of the data. In other

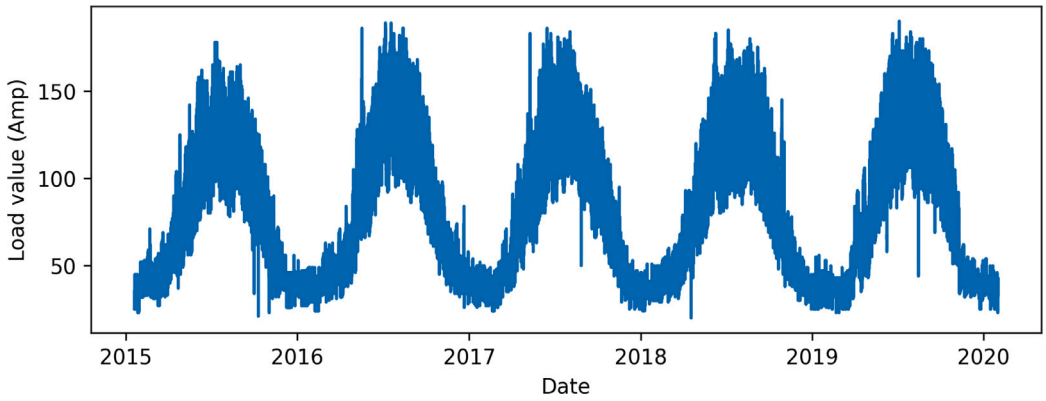
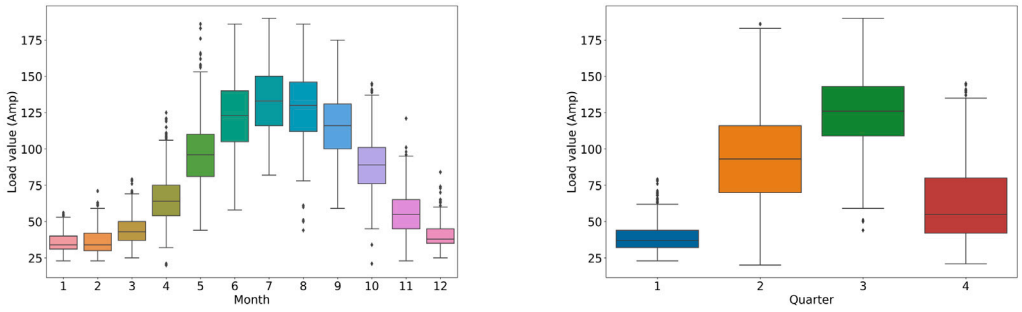


Fig. 3. Load profile of feeder 1 from 2015 to 2020. Load profiles of other feeders are similar to this.



(a) Box plot demonstrating relation between the load and months of the year

(b) Box plot of the load profile throughout different quarters

Fig. 4. Box plots unveiling the relationship between load value and months of the year, and the load profile across various quarters.

words, it contains the middle 50% of the data. Inside the box, there is a line that represents the median of the data, which gives an idea of the central tendency of the data within the specific month. The height of the box indicates how spread out the data is within this range. The data variability within a month is significantly less, highlighting the importance of adding a month as a separate feature.

Fig. 4(b) shows the box plot of the load profile throughout the different quarters. The x-axis of the plot is divided into four sections, each corresponding to a quarter of the year (Q1, Q2, Q3, Q4). Load consumption is quite significant in the third quarter. Also, variability in the third quarter is noticeably lower, whereas load consumption is comparatively less in the first and fourth quarters. The variability of the load profile in the first quarter is also very low. This motivated us to consider quarters as a feature for load forecasting.

We have considered the day of the week as a feature for training machine learning models. Fig. 5(a) illustrates the energy consumption profile for each day of the week. We can observe that the fifth and sixth days, Friday and Saturday, which were weekends in Dubai before, had high energy usage compared to other days. As the resolution of our data is every 30 min, we have also considered the hour a feature. The load profile varies based on the hour of the day, as depicted in Fig. 5(b). We can see that the high usage occurs between 11 a.m. and 8 p.m. We incorporate this hourly effect into our models by treating it as a feature. As we discussed before, temperature and humidity can affect the load value. The need for energy will rise during summers as the air conditioner usage will increase. Even a reduction in temperature can increase energy consumption with the use of heaters. However, in Dubai, the temperature never drops below 16 degrees Celsius, eliminating the need for a heater. The rise in energy usage with the temperature rise is shown in Fig. 6(a). Humidity is the amount of water vapor present in the atmosphere. As the humidity increases, the load usually increases. The graph in Fig. 6(b) shows that the high humidity value is associated with the winter season. Hence, fog contributed to the high humidity value. As a result, in our dataset, we can see an inverse relationship between humidity and load. Thus, we considered both temperature and humidity in our feature set.

We identified a correlation between each feature and the load using the Pearson correlation coefficient technique. This method assesses the degree of linear association between two data sets. The Pearson correlation coefficient, $\rho(X,Y)$, is defined as:

$$\rho(X,Y) = \frac{Cov(X,Y)}{\sigma_X \sigma_Y} \tag{1}$$

where,

cov = covariance of X and Y

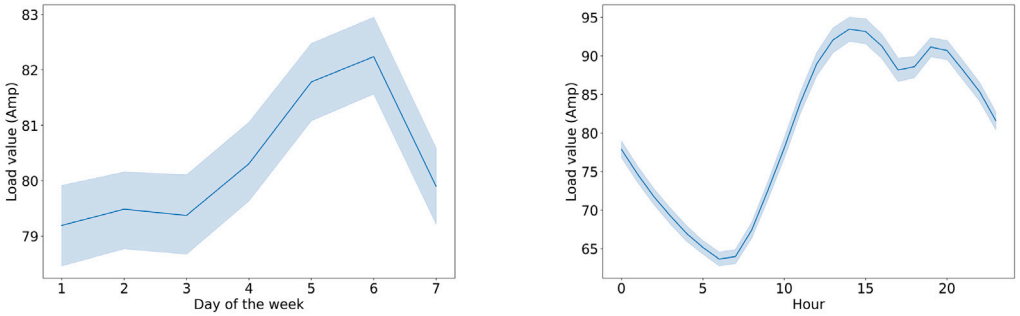
σ_x = standard deviation of X

σ_Y = standard deviation of Y

The correlation map among variables is shown in Fig. 7. We can interpret this figure as the correlation between temperature and load value is 0.75. The correlation plot for each feeder varies. However, in all feeders, we have observed a high correlation between load and temperature.

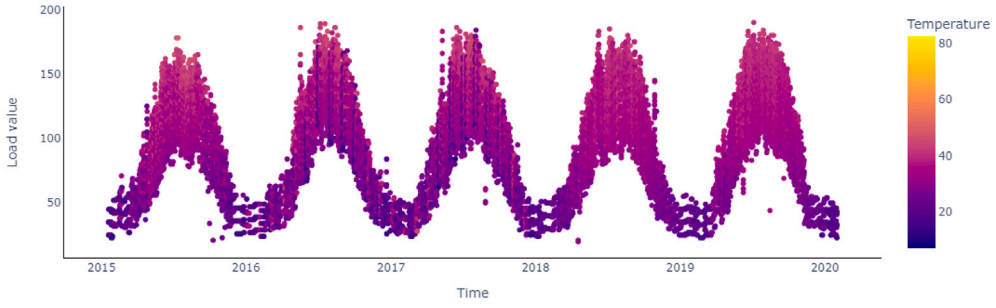
2.2. Model selection

This work has tested a library of machine-learning models for medium load forecasting for the feeder data. Out of all the tested models, the following proved good for medium-term forecasting accuracy with models like XGBoost [21], SGD Regressor [22], RandomForest [23], and LSTM [5,10,24]. Each model has its strength and is applicable in different applications. The selection of the model varies

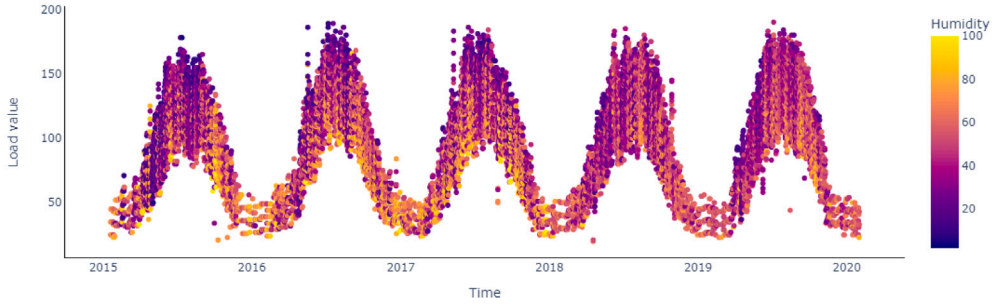


(a) Load varying with days (b) Load varying with hours in a day

Fig. 5. Variation of load profile with respect to day of the week and hour of the day.



a) Temperature dependence of the load.



b) Humidity dependence of load.

Fig. 6. Multivariate plot demonstrating the relationship between temperature and humidity with load consumption.

with the data in machine learning. In this study, we decided to proceed with XGBoost model after experimenting with aforementioned models. Reasons for selecting XGBoost model is discussed in Sections 3.1 and 4

2.3. An overview of the XGBoost model

XGBoost, defined as the extra gradient boost technique, is an ensemble learning technique in machine learning. This technique is a staple for many award-winning competitions and real-life applications. XGboost is a powerful and flexible machine learning technique commonly used in load forecasting [21,25]. The architecture of XGBoost model is shown in Fig. 8. XGBoost uses a boosting technique consisting of multiple-week learners, typically decision trees. It learns sequentially, where each learner will try to reduce the error residuals of the previous learners.

XGBoost employs a leaf-wise growth strategy, expanding the tree by choosing the split that minimizes loss. The optimization objective of XGBoost involves minimizing a loss function, which can be a combination of data loss terms and regularization loss. In our medium-term load forecasting, we are trying to predict the load value one year ahead. As our problem is a regression problem, the data loss term in the optimization term will either minimize the mean square error or maximize the R2 loss. The regularization term can be L1 (lasso) or L2 (ridge) or both, which avoids overfitting and aids in the model's generalization.

The objective function,L to minimize is,

$$L = Loss(y_i, \hat{y}_i) + 0.5 * \lambda_1 * \Sigma_{k=1}^{K}(w_k) + 0.5 * \lambda_2 * \Sigma_{k=1}^{K}(w_k)^2 \quad (2)$$

where $Loss(y_i, \hat{y}_i)$ is the mean square error loss which is defined as:

$$Loss(y_i, \hat{y}_i) = \frac{1}{n}\Sigma_{i=1}^{n}(y_i - \hat{y}_i)^2 \quad (3)$$

where y_i is the actual load value, $\hat{y}_i$ is the predicted load value. w_k are the weights of the model. λ_1 and λ_2 are the regularization parameters

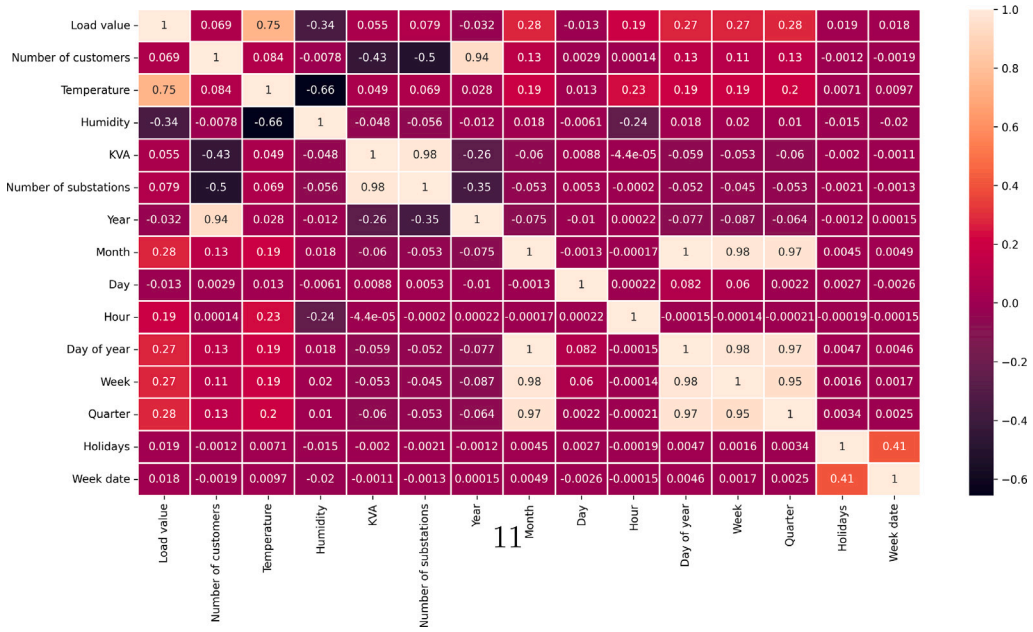


Fig. 7. Correlation plot for different features.

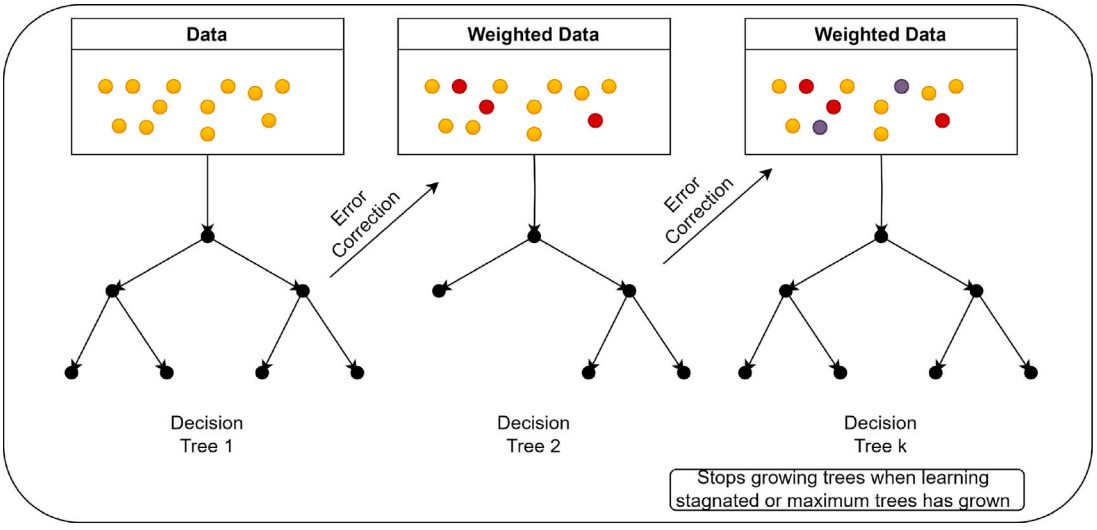


Fig. 8. Architecture of XGBoost model.

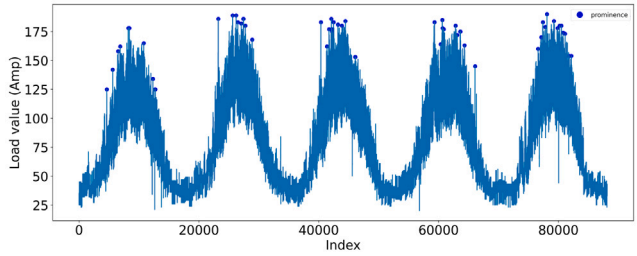
for L1 and L2 regularization, respectively. The model optimizes the objective function using the gradient descent optimization technique. XGBoost can also deal with missing data, which reduces the effort in preprocessing techniques and it is robust to outliers.

XGBoost model forecasts the medium-term load with half an hour resolution. However, designers must still craft these models specifically for peak accuracy prediction. Even if we can use the predicted values for identifying the peak and the corresponding peak accuracy, the model is not tuned for peak prediction. In the next section, we propose a Prominence-guided Weighted Peaks XGBoost Model(PWP-XGBoost) that integrates into our machine learning model to improve the peak accuracy along with one-year ahead load prediction with an hour resolution.

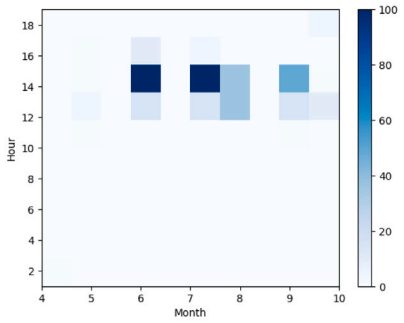
2.4. Prominence-guided weighted peaks model

Peak load prediction is crucial for distribution planners in the power utility sector, as it plays a vital role in efficient system design, resource allocation, and overall grid management. For power entities, predicting peak load for the future horizon is necessary to plan to meet the distribution demands of their areas. The peak load values occur at the peaks of summer time, especially in the MENA region. It is necessary to predict these peak values to avoid power shortage/outage. Existing ML models aim to get a high overall fit of the predicted values to the actual values, and these models do not focus on improving peak accuracy. Leveraging the advantage of the boosting technique of the XGBoost model, we combine it with a weighted peak approach to enhance the peak load prediction accuracy. We utilized the highest load consumption months and days of the year as added weights to the loss function of the XGBoost model. As the boosting technique involves the correction of weights in every iteration, these weights will incorporate the information on peak load consumption in the dataset. Hence, we developed a weighted peak model on top of the XGBoost model to improve the peak accuracy of the prediction results.

We used a prominence peak plot as shown in Fig. 9(a) to capture the highest consumption months and hours. The prominence of a peak is defined as the vertical distance between the peak and its lowest



(a) Load profile with peak loads circled



(b) Bivariate plot for months and hours

Fig. 9. (a) Load profile with peaks dotted using blue dots. The peaks are prominent during the summer months. (b) A bivariate histogram plot was used to evaluate the most occurring months and hours of the day with peak load value. It can be seen that months 6 and 7 and hours 12–16 have peak load values.

contour line. It indicates how much a peak is distinct from the surrounding baseline of the signal. We used this prominence peak plot as a data visualization tool for witnessing the significant and distinct peaks and thus captured the highest consumption months and hours. The months and hours having these peak values are plotted using a bi-variate histogram plot in Fig. 9(b). The dark blue color indicates that months 6 and 7 and hours 12–16 have the highest peak load values. These specific months and hours are assigned a higher weight value when compared to the regular months. The value of weight is a hyperparameter tuned for its optimum value. As discussed before, for the standard optimization algorithm of the machine learning regression problem, the mean square error is the loss function. For our custom model, we added more weights to the loss function corresponding to the locations where we expect high load values. These locations are obtained using Figs. 9(a) and 9(b), as explained above. We define this custom loss as prominence-guided peak loss. We can formulate it as,

$$\begin{aligned}\text{PWP-XGBoost loss}(y_i, \hat{y}_i) &= \frac{1}{n}\Sigma_{i=1}^{n}(y_i - \hat{y}_i)^2 + \frac{1}{n}\Sigma_{i=1}^{n}(y_i - \hat{y}_i)^2 \times w \times p \\ &= \frac{1}{n}\Sigma_{i=1}^{n}(y_i - \hat{y}_i)^2(1 + w \times p)\end{aligned} \tag{4}$$

where,

weight, $w \in \mathbb{R}$

peak, $p = 0|1$

We identified the value of w using hyperparameter tuning. p indicates the presence of the peak. If the peak is present in that month and hour, the p is 1; otherwise, the p is zero. Optimizing PWP-XGBoost loss aids in predicting the peak more accurately.

2.5. Model training and cross-validation

To achieve the best possible performance of the models, their parameters must be set to optimum values. The tuning of these parameters, better known as hyperparameter tuning, is performed using cross-validation (CV) techniques and functions that loop through various sets of values to finalize the set that provides the best performance of the model. CV refers to splitting the dataset into varying lengths of training and testing subsets and using varying sets of parameter values to evaluate the model's performance. We split the data set into five folds, and each fold (portion) is tuned through the train-validate split. As we conduct time series forecasting, the splitting is conducted sequentially to preserve the temporal order of the data, with the first portion of the data set taken as training and the latter as validating/testing. It is critical that the split cannot be performed randomly as our data is time-dependent and linear. As we are predicting one year ahead of the load forecast, the data from 2015–2019 is taken as training for the final model evaluation, and the data from 2019–2020 is taken as testing. This duration is variable for different feeders.

Table 1
Hyperparameters used for XGBoost model.

Hyperparameters	Obtained values after tuning
Number of trees	976
Tree depth	5
Subsample	0.46
Column Subsampling by Tree	0.70
Gamma	0.92
L1 regularization term	0.70
L2 regularization term	0.50

The hyperparameters that we tuned are the number of trees, the total number of trees added to the model, Tree depth, which decides the depth of each tree in the ensemble model, Subsample: fraction of training data randomly sampled from training data, Column Subsampling by Tree: Fraction of features (columns) randomly sampled for each tree, gamma: Minimum loss reduction required to make a further partition on a leaf node, l1 regularization term: adds penalty to the weights regularization term and L2: adds penalty to the squared weights.

We utilized Optuna, an optimization framework for hyperparameter tuning. It allowed us to search through hyperparameter space automatically and identify the hyperparameters that help in accurate load prediction. The table below depicts the hyperparameters obtained: (see Table 1).

2.6. Performance metrics

To evaluate the performance of the models for load prediction, we use three metrics: R^2 score, RMSE, and MAPE.

R^2 score, also known as the coefficient of determination, is a popular machine learning metric to judge the goodness of fit of the model to the data set. Its value ranges between 0 and 1, with 1 being an indication of 100% accuracy or a perfect model. In real-life situations, it is impossible to attain a score of 1, hence R^2 values closer to one are deemed excellent. The formula is:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{5}$$

RMSE, or Root Mean Squared Error, is the square root of the average squared difference between the actual and predicted load values. The formula is:

$$RMSE = \sqrt{\frac{1}{n}\Sigma_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{6}$$

MAPE, or Mean Absolute Percentage Error, is the average of the absolute ratio of the difference of predicted and true load value to the true load value. The formula is,

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \times 100\% \tag{7}$$

where,

y_i = actual load value

Table 2
Metric evaluation of the model performance for 5 feeders.

	R^2 score			
	RF	SGD	XGB	LSTM
Feeder 1	**0.95**	0.94	**0.95**	0.94
Feeder 2	0.94	**0.94**	0.92	0.93
Feeder 3	0.93	0.92	**0.94**	0.89
Feeder 4	**0.92**	0.71	**0.92**	0.91
Feeder 5	**0.93**	0.87	**0.93**	0.87
	RMSE			
	RF	SGD	XGB	LSTM
Feeder 1	9.16	10.72	9.21	**7.94**
Feeder 2	7.12	6.97	8.00	**5.76**
Feeder 3	9.79	10.21	**9.17**	9.45
Feeder 4	8.12	15.56	7.96	**6.79**
Feeder 5	**9.50**	12.96	9.78	9.62
	MAPE			
	RF	SGD	XGB	LSTM
Feeder 1	10.98	13.60	10.98	**10.55**
Feeder 2	12.92	13.45	14.20	**7.48**
Feeder 3	15.39	16.24	13.75	**12.42**
Feeder 4	12.62	21.72	12.79	**8.76**
Feeder 5	**10.18**	12.87	10.95	13.04
	Computational time in seconds			
	RF	SGD	XGB	LSTM
	15.79	0.045	4.68	366.74

Table 3
Comparison of peak accuracy obtained using XGBoost and PWP - XGBoost models.

	Peak accuracy	
	XGBoost	PWP - XGBoost
Feeder 1	85.2	90.3
Feeder 2	73.2	74.7
Feeder 3	92.5	94.6
Feeder 4	73.6	78.5
Feeder 5	89.09	89.55

$\hat{y}_i$ = predicted load value

$\bar{y}$ = mean of the actual load values

n = number of values

3. Results

This section explains the training and testing results of 4 different feeders using ML models RF, XGB, SGD, and LSTM. Their metric evaluations are shown to analyze the best-performing model. Furthermore, a plot for the predicted and true load values is plotted to show the performance visually. Finally, the metrics of the performance of the weighted peaks model plot are discussed to prove its improvement over the existing ML models.

3.1. Results across different machine learning models

Table 2 shows the R^2 score, RMSE, and MAPE for 5 feeders for the different ML models.

R^2 score of the XGB model is the highest for 4 out of 5 feeders. The RMSE and MAPE values are low for the LSTM model for the majority of the feeders. However, during training, LSTM models required hyperparameter tuning for each feeder to achieve acceptable performance. It lacked the generalization advantage of the XGB model. Furthermore, the computational time of XGB is 4 s, whereas the computational time for LSTM is 366.74 s. The RF and SGD models performed well. However, the overall performance of the XGB model is the best when compared to other models.

We plotted the testing dataset's forecasted and actual load values using the XGB model for feeder 1, as shown in Fig. 10. The true load values are shown in blue. The predicted load values are highlighted in red and superimposed over the true values of the testing dataset. The training dataset is from 2016–2018, and the testing dataset is from 2019–2020.

3.2. Results of our PWP - XGBoost model

The peak accuracy is calculated using the formula,

$$\text{Peak accuracy} = (1 - (\frac{|y_p - \hat{y}_p|}{y_p})) \times 100$$

where,

y_p = actual peak load value for a give year

$\hat{y}_p$ = predicted peak load value for a given year

For all 5 feeders, there has been an increase in the peak accuracy of the predicted load values. The increase is shown in Table 3 .Fig. 11 shows the increase in peak accuracy using PWP.

3.3. Explainability of model through feature importance

The feature importance graph is one of the tools for explaining the machine learning model, thus making it more transparent. It aids in visualizing the significance of different features in the model's prediction. By analyzing the feature importance graph, we can identify which feature had a more substantial effect on load forecasting. Fig. 12 depicts the feature importance graph of the XGBoost model for feeders 1, 2, 3, 4, and 5. In Fig. 12(a), we can observe that the one-year lag variable has a high effect on the model's prediction for feeder 1, whereas, in Fig. 12 (b) for feeder 2, the variable month has more influence on the prediction. For each feeder, variables aiding accurate prediction are different, validating the point of selecting all features.

4. Discussion

This work aimed to investigate the potential of Ml techniques for load forecasting and to integrate load peak prediction of feeders for planning and integrating energy resources for the following year. We considered five feeders in this work. We did data analysis and cleaning before model training. Multiple features like temperature, humidity, number of customers, number of substations, and KVA aid in machine learning prediction. We studied the correlation between each feature and the load value. Four ML models, Random Forest, SGDregressor, XGBoost, and LSTM, were trained and tested for all the feeders for medium-term load forecasting (1 year ahead). The XGBoost model and LSTM model gave the best results among the four machine learning models. RMSE and MAPE of the LSTM model are less than other models for all feeders. However, the LSTM model is very data-specific, requiring hyperparameter tuning for each feeder, and if the data is less, the LSTM model fails. Also, the performance of the XGBoost model was better than that of LSTM when the data was less. XGBoost model has achieved good results for a one-year time horizon, with an R^2 score of 0.9333 (closer to 1 means a more accurate model). Also, the training and inference time taken by the XGBoost model is comparatively much less than that of LSTM and other models. Also, we can execute the XGBoost model on a normal CPU, whereas we need GPUs to run the LSTM model. XGBoost model is more explainable than the deep learning models like LSTM. Also, the XGBoost model is easy to implement, and thus, we proceed with the XGBoost model for further analysis. We proposed a prominence-guided peak prediction model using XGBoost to aid in the power planning for the following year. We have identified the peak locations using a 2D histogram across months and times for

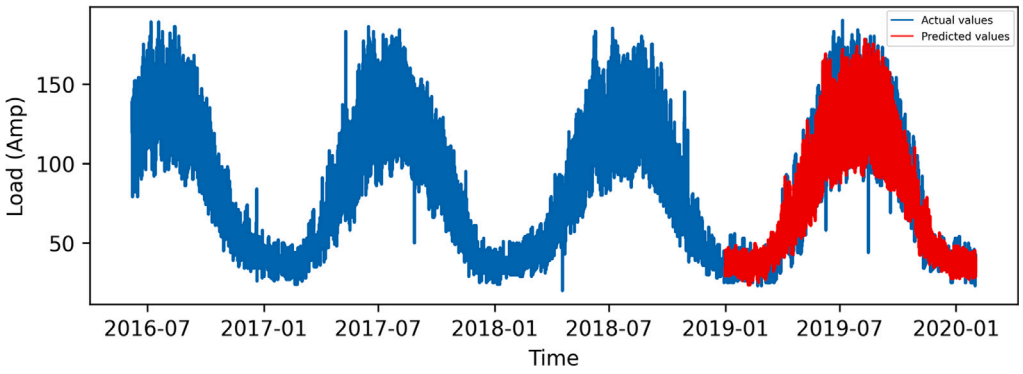


Fig. 10. Plot depicting the actual and predicted values obtained on feeder 1 using XGB model.

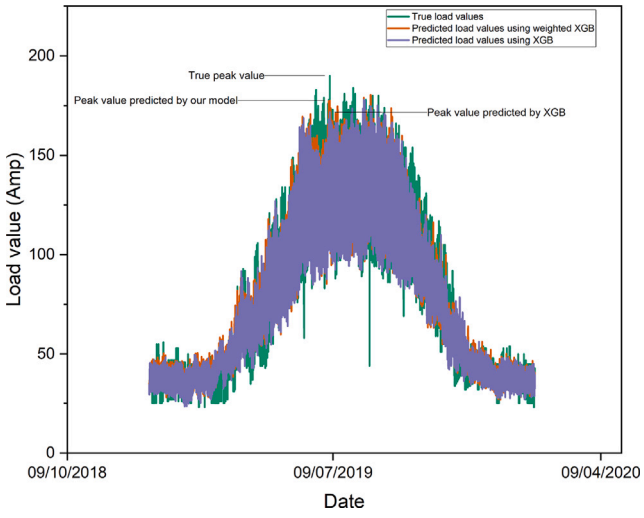


Fig. 11. Figure shows three plots for the testing dataset of feeder 1. The green plot is the actual load values, the orange plot is the values predicted by the PWP - XGBoost model, and the purple is the values predicted by the XGBoost model. The peak accuracy of XGBoost and PWP - XGBoost models are compared, showing an increase of 5% in accuracy.

prominence peaks and implied a weighted peak loss using XGBoost. It helped us to improve the peak prediction accuracy between 0.5% to 5%. Even a 1% improvement in peak prediction accuracy can have a high economic impact [26]. The proposed peak prediction algorithm can be easily integrated with the loss function of machine learning models like LSTM or Random Forest if required.

5. Conclusions

The implementation of the PWP-XGBoost model for medium-term feeder load forecasting and enhanced peak accuracy prediction presents a promising approach with several notable advantages. By integrating historical load data, weather patterns, and exogenous variables, the model demonstrates a capacity to capture complex load behaviors and adapt to changing conditions. We compared the results with different machine learning models like XGBoost, RandomForest, and LSTM for medium-term load forecasting. Based on the results, we identified that the XGBoost model works better for the dataset used in this study. The model facilitates proactive measures to optimize energy distribution, mitigate grid instability, and reduce operational costs by accurately predicting peak load periods. In essence, the PWP-XGBoost model represents a significant step forward in the field of load forecasting, offering utilities and stakeholders a powerful tool to navigate the complexities of modern energy systems and pave the way for a more sustainable and resilient future.

While the model is designed to adapt, it may struggle with rapid changes in load behavior not seen during training. This necessitates frequent retraining and model updates, and integrating new data sources or types of data into the existing model can be challenging and may require significant modifications to the data preprocessing and feature engineering pipelines. By addressing these identified weaknesses and building on the strengths of the PWP-XGBoost model, we can further enhance its utility and effectiveness, paving the way for more accurate, efficient, and reliable load forecasting in modern energy systems.

As an extension to this work, we will explore developing a model that represents as many feeders as possible by incorporating unsupervised machine learning into the data learning process. We will also investigate using Generative AI models to support this process. Moreover, the next phase of this work will leverage the Operational Machine Learning (OpML) framework to integrate our research into an operational system deployable by businesses. This will involve fully integrating existing data sources, utilizing existing infrastructure, and enhancing the user experience with the processes developed in this work.

CRediT authorship contribution statement

Ansu Mathew: Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Conceptualization. **Rufaidah Chikte:** Writing – original draft, Visualization, Methodology, Investigation. **Sajan K. Sadanandan:** Writing – review & editing, Writing – original draft, Supervision. **Sayed Abdelaziz:** Data curation. **Shoaib Ijaz:** Data curation. **Tareg Ghaoud:** Project administration, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Ansu Mathew reports administrative support was provided by Dubai Electricity and Water Authority PJSC. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

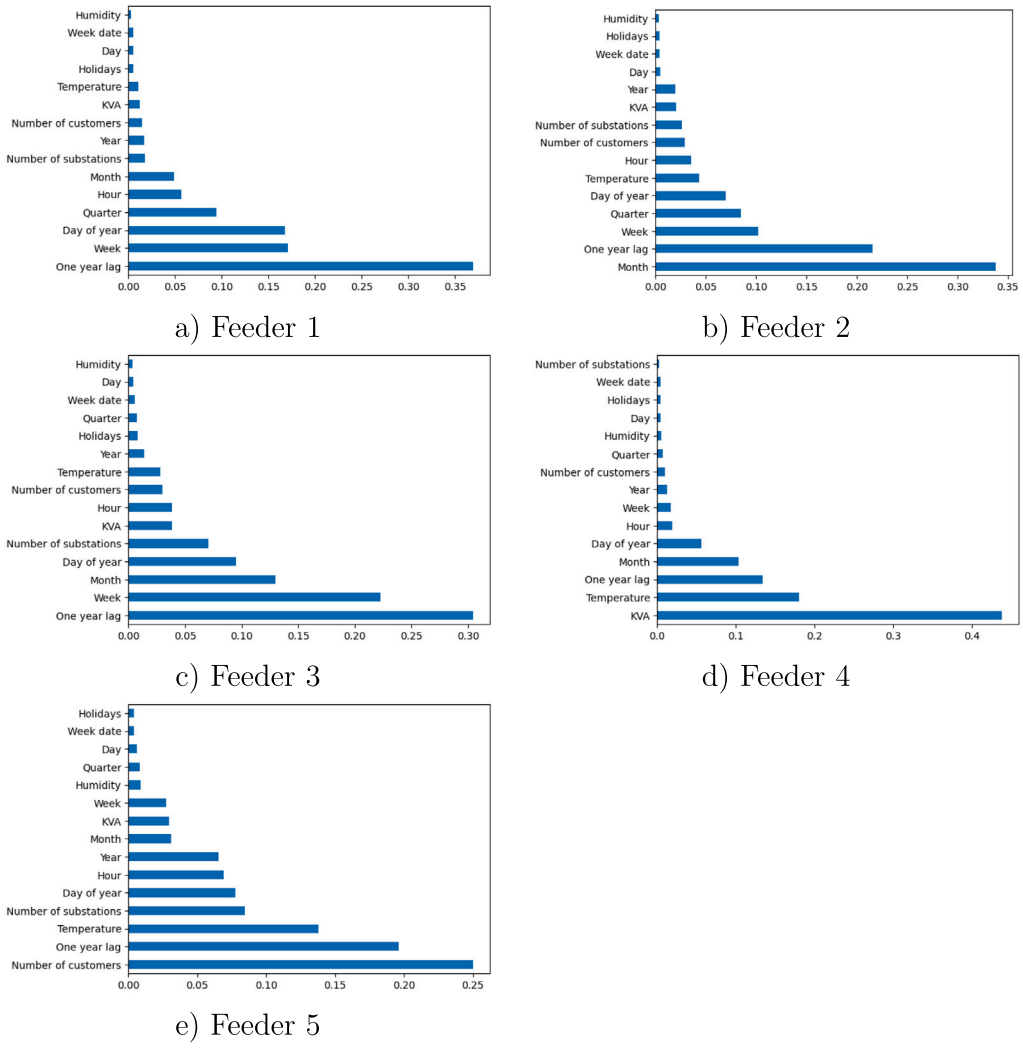


Fig. 12. Feature importance plots.

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